Comparative Analysis of Optimization Algorithms and the Effects of Coupling Hedging Rules in Reservoir Operations

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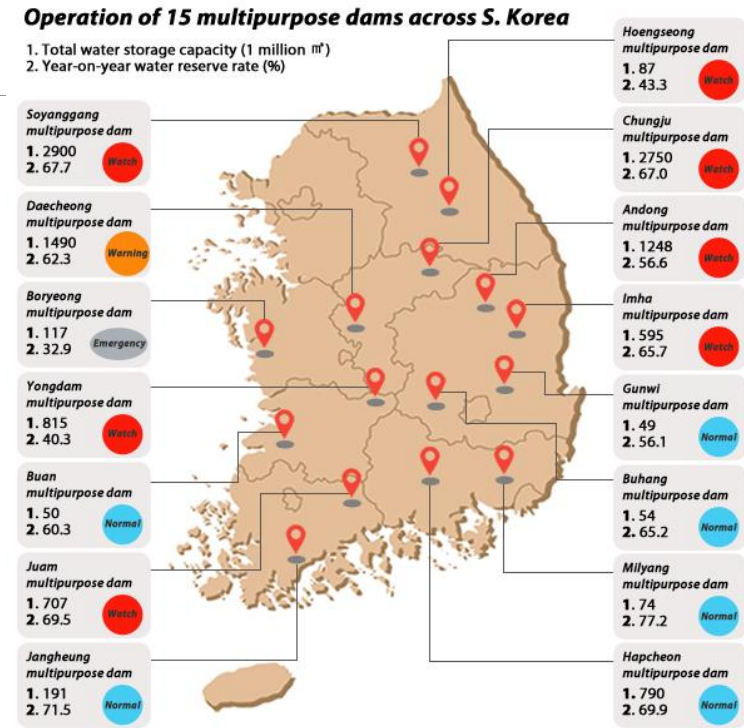
Recent multi-year droughts in Korea

Dried-up dams in S. Korea amid prolonged drought

(MBN, 2015)

Chang Young-seok 기자 | 입력 : 2015.11.08 14:38:39 수정 : 2015.11.10 13:04:52 0



Korea suffers worsening drought

(Korea Herald, 2015)

By Korea Herald Published : Jun 16, 2015 - 20:44 Updated : Jun 16, 2015 - 21:00

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Fig. 1. Articles regarding the recent multi-year droughts in Korea

Real-world policy in Reservoir Operations

❖ Real-World Policy: Zone-based Hedging Rule

- Discretizes the entire range of storage into Zones (Fig. 2.2)
- Release decisions are determined according to the "Zone" the current storage belongs to
- Selected as the main operating policy in all Korean multipurpose reservoirs (currently, 20 multi-purpose dams each with different curves)

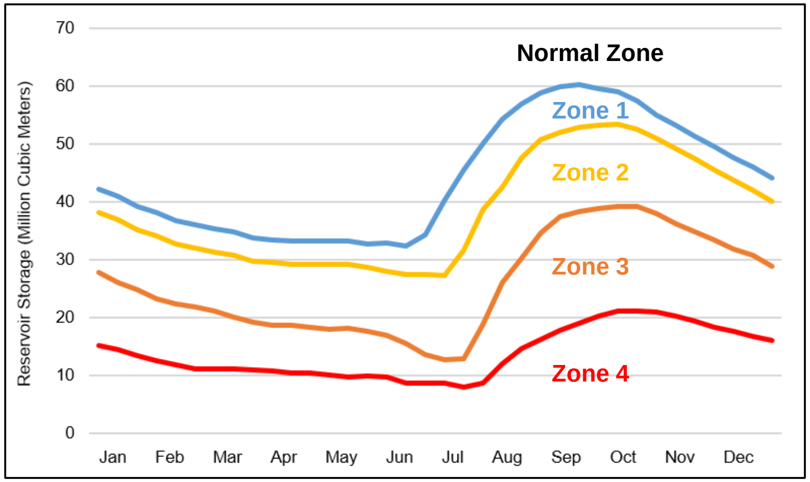

Fig. 2. Actual Rule Curves of Boryeong Reservoir, Korea

Table 1. Ratios implemented during hedging

Zone Name	Municipal & Industrial	Irrigation	Environmental
Normal Zone & Zone 1 (Attention Zone)	100%*	100%*	100%*
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Zone 4 (Critical Zone)	Up to 80%	70%	0%

* Release exceeding demand is possible in Normal Zones

Key Research Questions

- How does **different optimization models or their formulations** affect both the **quality and the diversity** of solutions in reservoir operation problems?

- How about when the reservoir was operated with the **status quo (SQ)** rules? How do they perform under both drought and normal conditions compared to optimal solutions?

- How does the **implementation of SQ rules with the solutions from optimization algorithms** improve (or degrade) the performance under extreme drought conditions?

Flow Chart of Overall Research

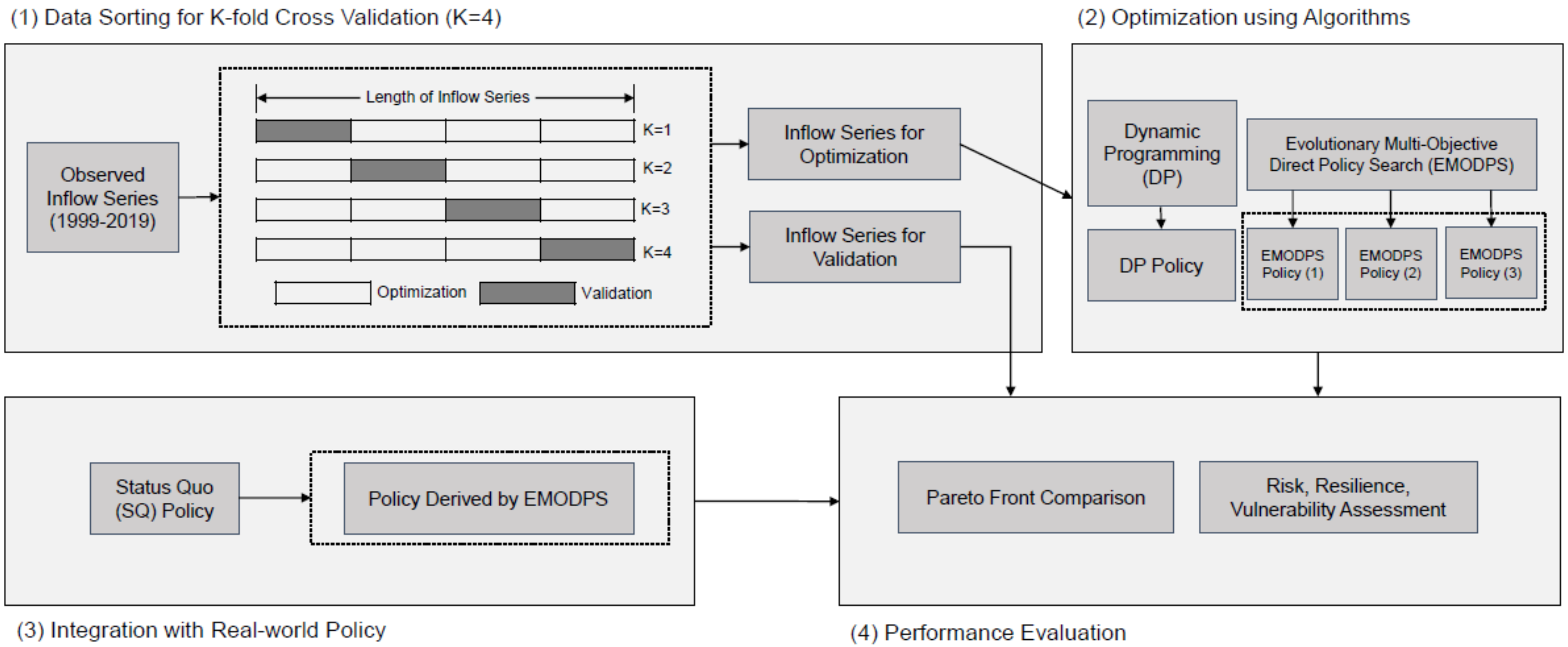


Fig. 3. Overall Flow Chart of this Research

❖ Keywords: Reservoir operation, Dynamic programming, Evolutionary multi-objective direct policy search, Zone-based hedging rule

Optimization Techniques for RO

❖ Selected Optimization Algorithms

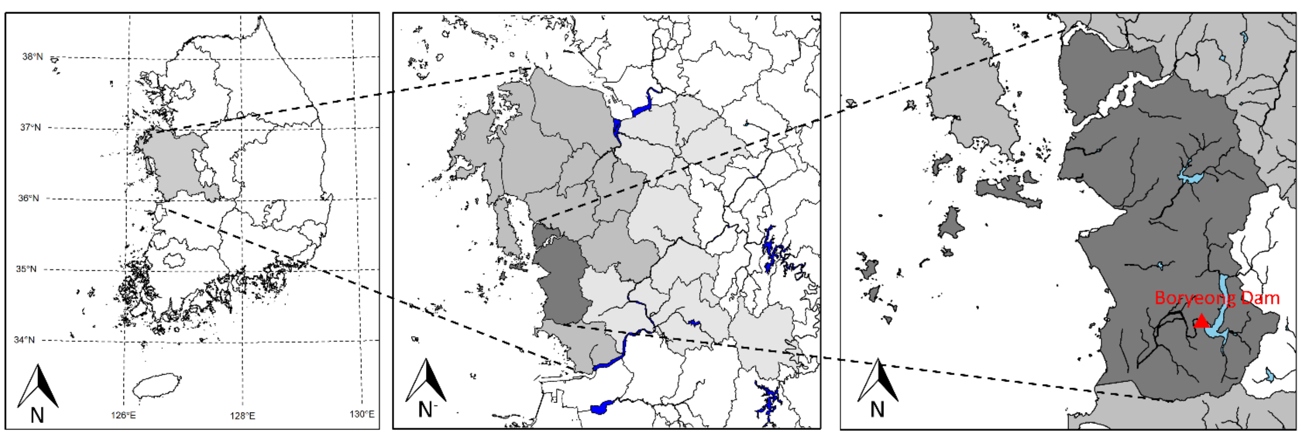
- Dynamic Programming
  - Discretizes the entire range of state, decision, and disturbance variables for iterative calculation that lead to the steady-state solution
  - Deterministic and stochastic versions of the DP models depending on the available information
- Evolutionary Multi-Objective Direct Policy Search **(EMODPS)**
  - **Direct Policy Search (DPS)** (Rosenstein and Barto, 2001)
    - A type of parameterization-simulation-optimization (PSO) approach in which the decision variable directly operates in the policy space
  - Nonlinear approximating networks: **Radial Basis Function (RBFs)**
    - Approximating function that determines the value of the release decisions
  - **MOEAs** during the solution searching process
    - Iterative search algorithm that evolve a Pareto-approximate set of solutions in a multi-objective problem

Case Study: Boryeong Multipurpose Reservoir

❖ Watershed Area: 2,860 km^2

❖ Water Supply Districts (8, Light Gray)
Boryeong (Dark Gray), Seosan, Dangjin, Seocheon, Cheongyang, Hongseong, Yesan, Taean

❖ Water Use: 106.6 MCM/year (Reservoir Capacity: 116.9 MCM)
(**Municipal & Industrial water demand: 84%,**
Agricultural water demand: 5%, Environmental water demand: 11%)



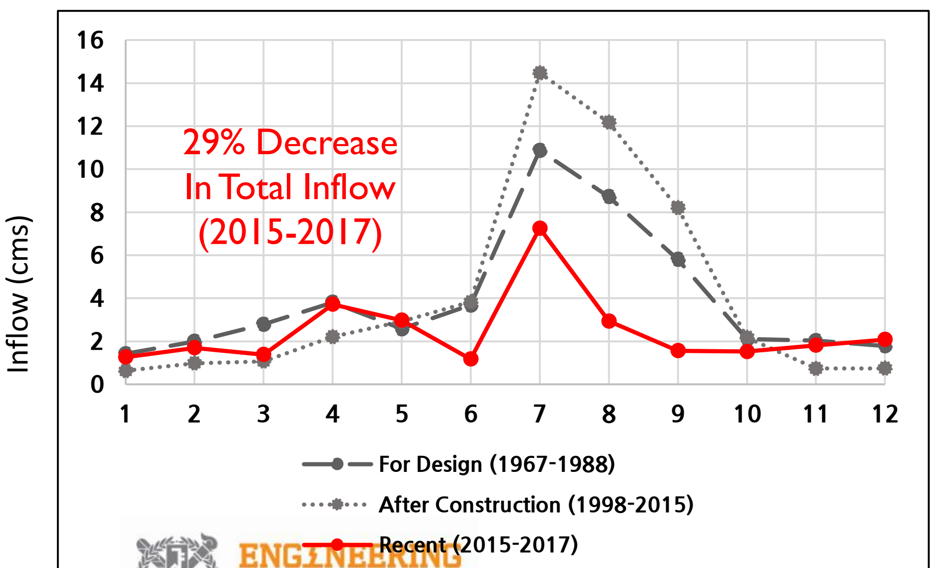
Republic of Korea | South Chungcheong Province | Boryeong City

Fig. 4. Map of the study basin

Case Study: Boryeong Multipurpose Reservoir

❖ Recent Multi-year Drought in Boryeong Dam Basin (2014-2019)

- ◆ Consecutive decrease in the total annual precipitation (6 water years)
- ◆ Both structural/non-structural measures to mitigate the effects from the long-lasting drought
  - Conduit Construction Project (2016-2017, $62,500,000+$2,600,000/year)
  - Enforcement of Pressure Reduction in Water Supply Systems (127 days)



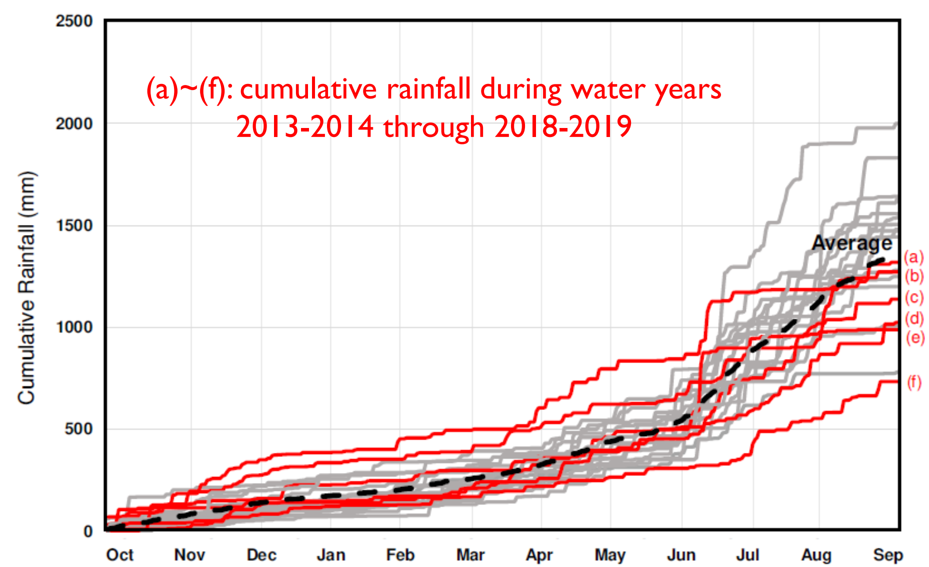


Fig. 5. Graphs of recent inflow decrease of Boryeong Dam

Problem Formulation: Objectives

❖ Reservoir Problem with Conflicting Objectives: J^{supply} vs J^{demand}

(both objectives to be minimized)

◆ J^{supply}: measures the magnitude of failure from the supply side

$$J^{supply} = \frac{1}{H}\sum_{t=1}^{H}\max\left(\frac{S_t^{target} - S_t}{S_t^{target}}, 0\right)\times 100(\%)$$

◆ J^{demand}: measures the magnitude of failure from the demand side

$$J^{demand} = \frac{1}{H}\sum_{t=1}^{H}\max\left(\frac{D_t - R_t}{D_t}, 0\right)\times 100(\%)$$

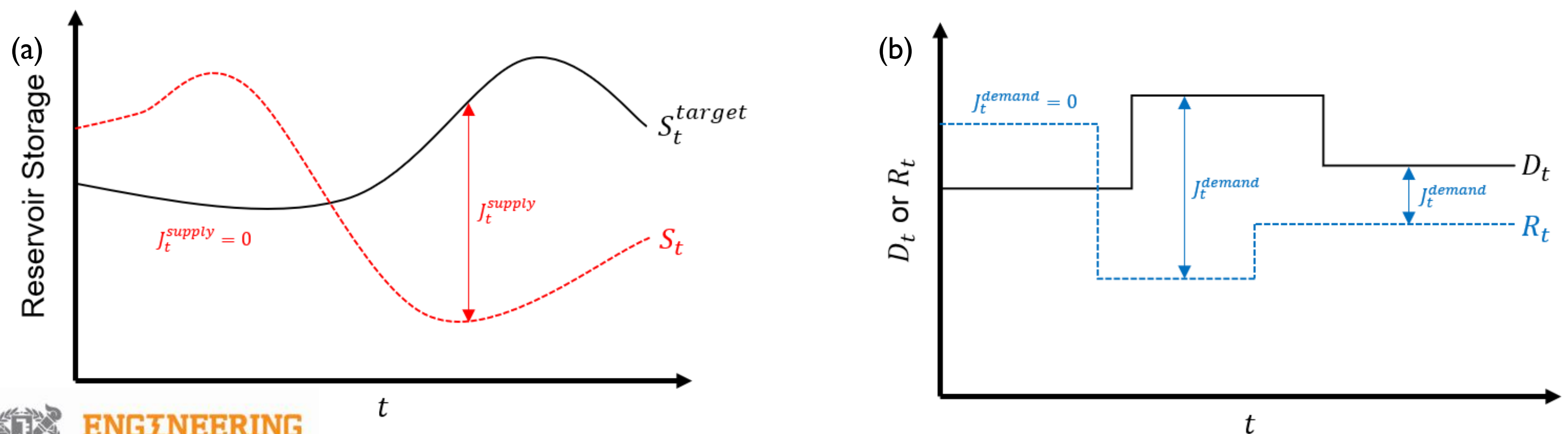


Fig. 6. Illustration of (a) J^{supply} and (b) J^{demand}

Problem Formulation: Optimization Models

❖ 2-Obj. Optimization Models for Optimal Reservoir Operations

◆ Dynamic Programming-based Models with different information

- Deterministic DP-Perfect (DDP-Perfect)
- Deterministic DP-Monthly Average (DDP-Average)
- Stochastic DP (SDP): Assuming lognormal distribution of inflows

$$f^{n}{}_{opt}(S_t, Q_t) = \max_{R^*_t}\left[Z_t(S_t, Q_t, R_t) + \underset{Q_{t+1}|Q_t}{E}\, f^{n-1}{}_{opt}(S_{t+1}, Q_{t+1}) \right] \qquad Z_t(\bullet) = w_1 \times J_t^{supply} + w_2 \times J_t^{demand}$$

◆ Evolutionary Multi-Objective Direct Policy Search (EMODPS) Models with different RBF shapes & number of parameters

- EMODPS-Linear: Linearly shaped RBF with 4 policy parameters
- EMODPS-Gaussian: Gaussian shaped RBF with 3 policy parameters
- **EMODPS-Root**: Root shaped RBF with 6 policy parameters (j=2)

$$R_t = \min\left(\max\left(\sum_{i=1}^{j} \left(w_i \times \left| D_t - \frac{S_t - d_i}{r_i} \right|^{0.5} \right), R_{\max} \right), R_{\min} \right)$$

Problem Formulation: Simulation

❖ K-fold Cross Validation (CV) Formulation

Table 2. Specific formulations for each subgroup by K-fold Cross Validation

K	Optimization Period (15 years)	Validation Period (5 years)	Average Inflow during Validation Period (m^3/s)	Starting Storage (Million cubic meters, date)
1	October 2004 - September 2019	October 1999 - September 2004	4.64	99.597 (Sep-30, 1999)
2	October 1999 - September 2004, October 2009 - September 2019	October 2004 - September 2009	3.65	89.978 (Sep-30, 2004)
3	October 1999 - September 2009, October 2014 - September 2019,	October 2009 - September 2014	4.39	64.365 (Sep-30, 2009)
4	October 1999 - September 2014	October 2014 - September 2019	2.72	47.364 (Sep-30, 2014)

Drought period with low starting storage (K=4)

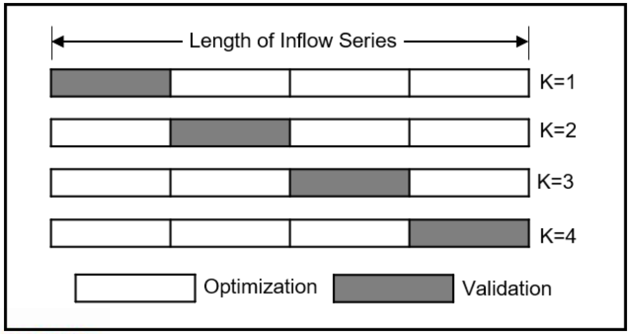


Fig. 7. Concept of K-fold CV

Results: Comparison among Models

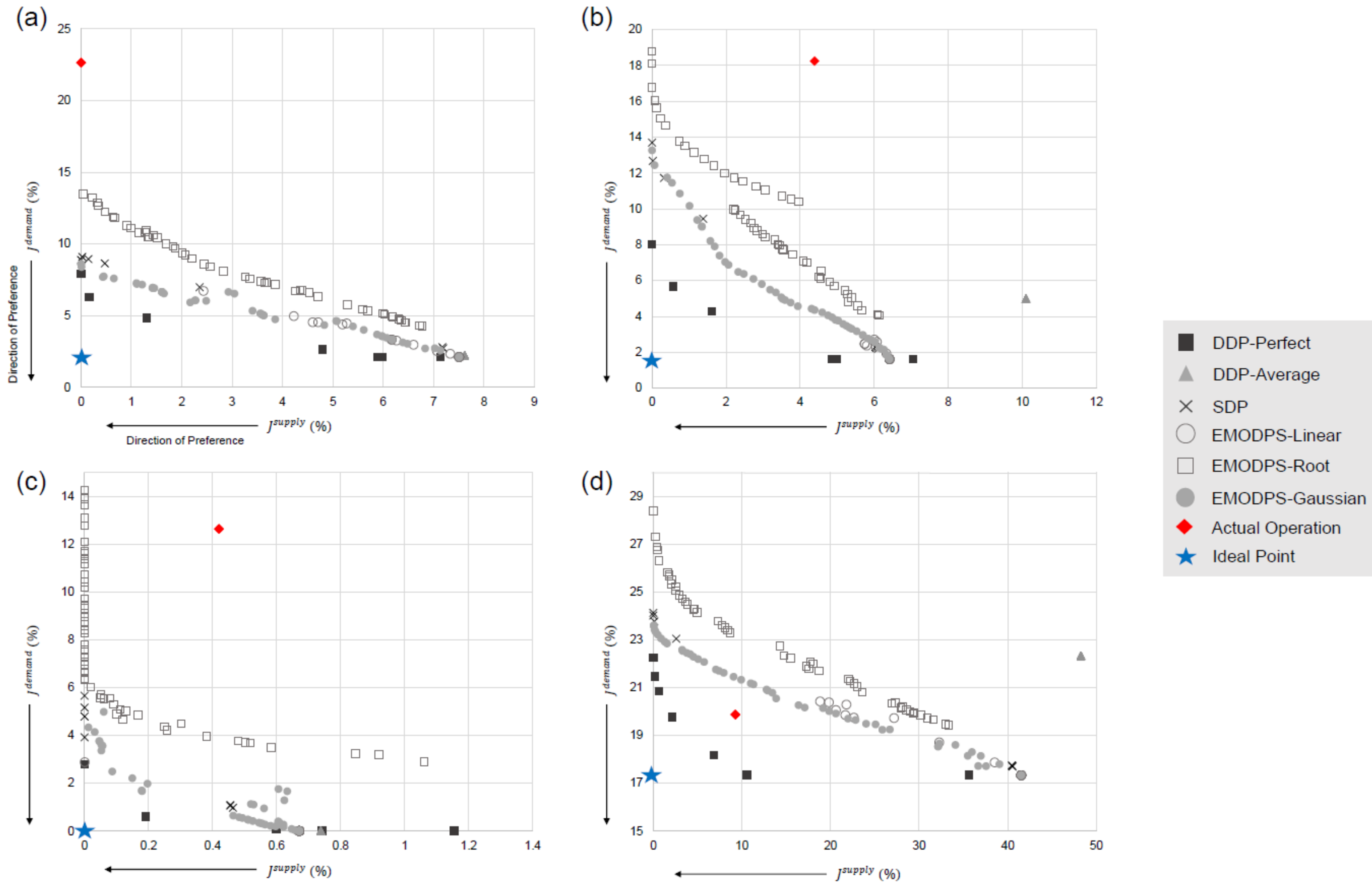


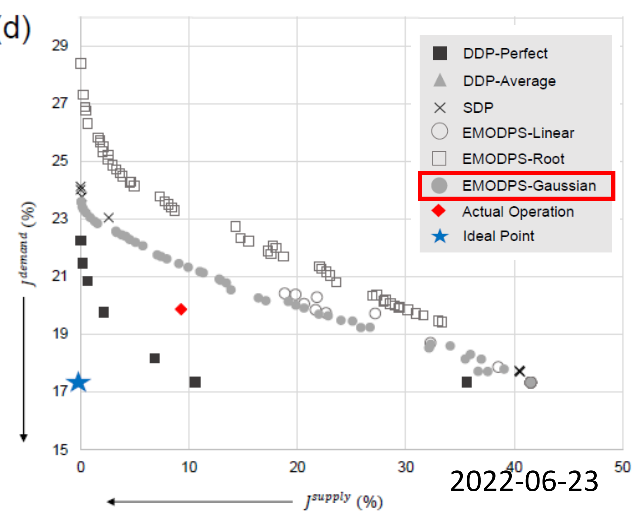
Fig. 8. Pareto Fronts of the solution sets derived by each subgroup in K-fold CV
(a) Subgroup K=1, (b) Subgroup K=2, (c) Subgroup K=3, and (d) Subgroup K=4

Results: Comparison among Models

❖ Insights from comparison among optimization models

- ◆ Among the EMODPS models, EMODPS-Gaussian model (●) with least number of parameters (3) performed best among all other models
  - Shape of the approximating function is more important than the number of parameters in this case
- ◆ Solutions from the SDP model (X) performed as well as the EMODPS-Gaussian model (●) in most of the subgroups, but they failed in terms of solution diversity (due to commensuration of objectives)
- ◆ Policy searching closed-loop models (SDP & EMODPS models) were more flexible than the deterministic open-loop models (DDP-Average, ▲)
- ◆ Actual operation based on the SQ policy (◆) based on zone-based hedging rules hold value under drought conditions (K=4)

(d)



Results: Coupling of Real-world Policy

❖ During simulation, real-world reservoir operating policy is coupled with the best performing optimization model: **EMODPS-Gaussian**

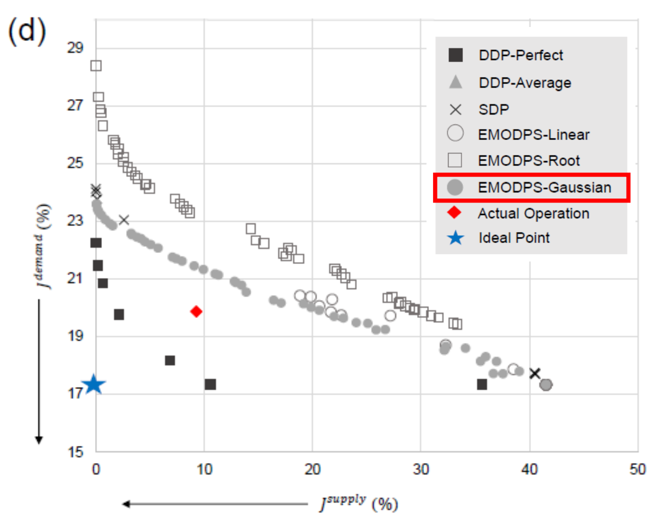
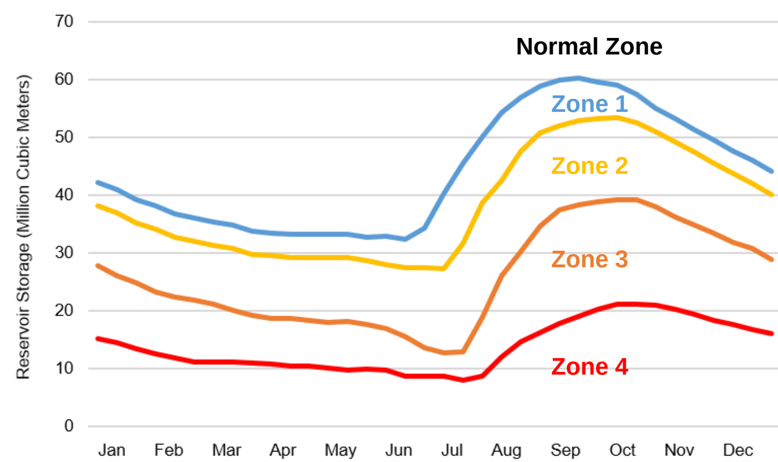


Fig. 8(d). Pareto Fronts of the solution sets derived from Subgroup K=4



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Results: Coupling of Real-world Policy

❖ Selected Performance Indices

- ◆ Risk, Resiliency, and vulnerability (Hashimoto et al. 1982)
  - Risk (frequency): How frequent was the failure?
  - Resiliency (duration): How fast does the system come back from failure?
  - Vulnerability (magnitude): How big is the magnitude of the failure?

- ◆ Evaluation of performance both from the Supply & Demand side (Definition of Failures)
  - Supply side (= J^{supply}): $S_t < S_t^{target}$
  - Demand side: $R_t < D_t^{M\&I}$ or $R_t < D_t^{Agr}$

Results: Coupling of Real-world Policy

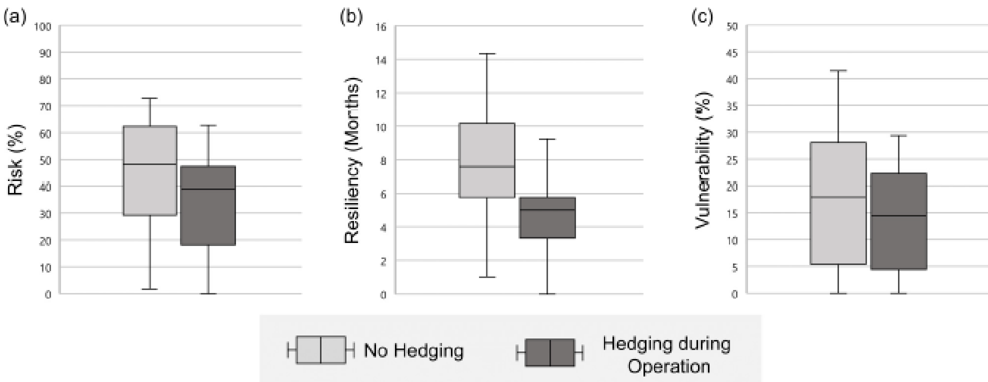

Fig. 10. Performance indices measured from the supply side: (a) risk, (b) resiliency, and (c) vulnerability

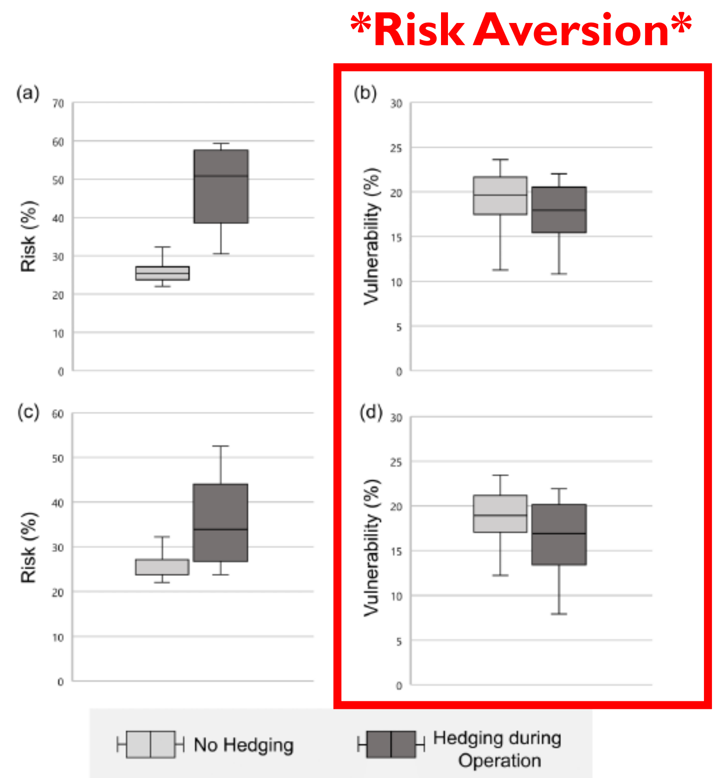

Fig. 11. Performance indices measured from the demand side: (a) risk of water for A uses, (b) vulnerability of water for A uses, (c) risk of water for M&I uses, (d) vulnerability of water for M&I uses,

Conclusion: Take-home Messages

❖ Under normal conditions, solutions from optimization models surpassed the results from actual operations

◆ There are definitely potential room for improvements with the introduction of optimization models!

❖ Among the optimization models, solutions from the EMODPS models dominated those from the DP models both in terms of proximity to the ideal point and the variety of solutions across the Pareto front

❖ Coupling of zone-based hedging rule with the solutions from the optimization models resulted in outcomes favorable to risk-averse real-world water users (may hold promise)

Related Publication

For More…

❖ Kim, G. J. and Kim, Y.-O.* "How does the coupling of real-world policies with optimization models expand the practicality of solutions in reservoir operation problems?" *Water Resources Management.* (in press)

Water Resources Management
https://doi.org/10.1007/s11269-021-02862-y

Check for updates

How Does the Coupling of Real-World Policies with Optimization Models Expand the Practicality of Solutions in Reservoir Operation Problems?

Gi Joo Kim[1] · Young-Oh Kim[1]

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Thank You!
(Boryeong-si, 2020.05.)